

Youngtaek Oh

Ph.D. Student, Korea Advanced Institute of Science and Technology (KAIST)

✉ youngtaek.oh@kaist.ac.kr <https://ytaek-oh.github.io> [in LinkedIn](#) [Github](#) [Google Scholar](#)

Research Interest

My research centers on developing data-efficient and robust recognition systems that address limitations and biases in image and video data. I am currently focusing on enhancing vision-language models to improve compositional reasoning and multi-modal understanding for more robust and comprehensive recognition systems.

Keywords: Data-Efficient Learning, Multi-Modal Learning, Vision-Language Models

Education

Korea Advanced Institute of Science and Technology (KAIST)	Daejeon, South Korea
Ph.D. in Electrical Engineering	Sep. 2021 – Present
◦ Co-Advisors: Prof. In So Kweon and Prof. Junmo Kim	
M.S. in Electrical Engineering	Mar. 2019 – Feb. 2021
◦ Advisor: Prof. In So Kweon	
Korea University	Seoul, South Korea
B.S. in Electrical Engineering	Mar. 2015 – Feb. 2019
◦ GPA: 4.40/4.50	

Publications (*equally contributed authors; †corresponding authors)

Workshop and Preprints

- [W2] [Exploring the Spectrum of Visio-Linguistic Compositionality and Recognition](#)
Youngtaek Oh, Pyunghwan Ahn, Jinhyung Kim, Gwangmo Song, Soonyoung Lee[†], In So Kweon[†], Junmo Kim[†]
 in **CVPRW 2024**: ‘What is Next in Multimodal Foundation Models?’ (**MMFM**) Workshop
- [W1] [Technical Report: Retrieval-based Data Discovery and Fusion for Zero-shot Image Captioning](#)
Youngtaek Oh, Jae Won Cho, Dong-Jin Kim, In So Kweon[†], Junmo Kim[†]
 in **CVPRW 2023**: ‘New Frontiers for Zero-shot Image Captioning Evaluation’ (**NICE**) Workshop
 2nd place in Zero-Shot Image Captioning Challenge in NICE Workshop

Peer-Reviewed Conferences and Journals

- [C6] [Preserving Multi-Modal Capabilities of Pre-trained VLMs for Improving Vision-Linguistic Compositionality](#)
Youngtaek Oh, Jae Won Cho, Dong-Jin Kim, In So Kweon[†], Junmo Kim[†]
 in **EMNLP 2024**: Conference on Empirical Methods in Natural Language Processing
 Oral presentation
- [J1] [Empirical study on using Adapters for debiased Visual Question Answering](#)
 Jae Won Cho, Dawit Mureja Argaw, **Youngtaek Oh**, Dong-Jin Kim, In So Kweon
 in **CVIU 2023**: Computer Vision and Image Understanding (IF=4.3)
- [C5] [Self-Sufficient Framework for Continuous Sign Language Recognition](#)
 Youngjoon Jang, **Youngtaek Oh**, Jae Won Cho, Myungchul Kim, Dong-Jin Kim, In So Kweon, Joon Son Chung
 in **ICASSP 2023**: IEEE International Conference on Acoustics, Speech and Signal Processing
 Oral presentation, Top 3% Paper Recognition
- [C4] [Signing Outside the Studio: Benchmarking Background Robustness for Continuous Sign Language Recognition](#)
 Youngjoon Jang, **Youngtaek Oh**, Jae Won Cho, Dong-Jin Kim, Joon Son Chung, In So Kweon
 in **BMVC 2022**: British Machine Vision Conference

- [C3] [DASO: Distribution-Aware Semantics-Oriented Pseudo-Label for Imbalanced Semi-Supervised Learning](#)
Youngtaek Oh, Dong-Jin Kim, In So Kweon
in **CVPR 2022**: IEEE/CVF Conference on Computer Vision and Pattern Recognition
Finalist, Qualcomm Innovation Fellowship Korea, 2022
- [C2] [KSL-Guide: A Large-scale Korean Sign Language Dataset Including Interrogative Sentences for Guiding the Deaf and Hard-of-Hearing](#)
Soomin Ham, Kibaek Park, Youngjoon Jang, **Youngtaek Oh**, Seokmin Yun, Sukwon Yoon, Chang Jo Kim, Han-Mu Park, In So Kweon
in **FG 2021**: IEEE International Conference on Automatic Face and Gesture Recognition
- [C1] [SideGuide: A Large-scale Sidewalk Dataset for Guiding Impaired People](#)
Kibaek Park*, **Youngtaek Oh***, Soomin Ham*, Kyungdon Joo*, Hyokyoung Kim, Hyoyoung Kum, In So Kweon
in **IROS 2020**: International Conference on Intelligent Robots and Systems

Work Experience

Research Intern	Sep. 2023 – Feb. 2024
LG AI Research	Seoul, South Korea
◦ Worked on vision-language compositionality; Outcome: Paper [W2] and Software [S3]	
Graduated Researcher	Mar. 2021 – Aug. 2021
Korea Advanced Institute of Science and Technology (KAIST)	Daejeon, South Korea

Projects

Developing and Demonstrating Innovative Products Based on Public Demand	Nov. 2021 – Present
funded by the Ministry of Science and ICT (MSIT), Korea	
◦ Consortium: KAIST, Hanbat National University, Miru Systems, Hanulsoft, Datamaker, Daejeon Transportation	
◦ Objective: Develop a real-time system for masking and unmasking personal information in public CCTV services.	
◦ Role: Designed a deep steganography algorithm to conceal and reveal personal information (face and license plate).	
Korean Sign Language Dataset for AI Interpretation	May 2020 – Dec. 2020
funded by the National Information Society Agency (NIA), Korea	
◦ Consortium: KAIST, Korea Association of the Deaf, Testworks, Korea Nazarene University, EQ4ALL	
◦ Objective: Establish a large-scale Korean Sign Language dataset for real-world use in sign language recognition.	
◦ Role: Developed a continuous sign language recognition model compatible with Korean Sign Language datasets.	
◦ Outcome: Publications ([C2], [C4], [C5]), dataset (KSL-Guide ↗ , <i>available to Koreans only</i>)	
Korean Sidewalk Image Dataset for AI Assistance to the Visually Impaired	May 2019 – Dec. 2019
funded by the National Information Society Agency (NIA), Korea	
◦ Consortium: KAIST, Korea Spinal Cord Injury Association, Testworks, SelectStar, DTWORESOURCE	
◦ Objective: Construct a dataset for sidewalk environments to enhance mobility rights for individuals with disabilities.	
◦ Role: Designed the sidewalk obstacle dataset and validated it using object recognition models.	
◦ Outcome: Publications ([C1]), dataset (SideGuide ↗ , <i>available upon request: application ↗</i>)	

Honors and Awards

Top Reviewer, NeurIPS 2024	Nov. 2024
Top 3% Recognition Certificates, ICASSP 2023	June 2023
Second place in the 2023 NICE Challenge at the NICE Workshop, CVPR 2023 (\$5,000)	May 2023
Finalist, Qualcomm Innovation Fellowship Korea	Nov. 2022
Outstanding Reviewer, ECCV 2022	Oct. 2022
National Scholarship for Science and Engineering (<i>full tuition, merit-based</i>)	Mar. 2017 – Feb. 2019

Softwares

Open-Sourced PyTorch Implementations

- [S4]  **fsc-clip**: Implementations of training CLIP models with hard negative captions, related to [W2], [C6].
 - Supports training on 3 image-text datasets with unified evaluation using **vl_compo**.
- [S3]  **vl_compo**: A unified evaluation toolkit for compositional reasoning and multi-modal tasks, related to [W2], [C6].
 - Incorporates a wide range of models and benchmarks for evaluation, continuously evolving to stay up-to-date.
- [S2]  **retriever**: An implementation of the second-place solution in the 2023 NICE Challenge, related to [W1].
 - Enhances BLIP-2 with retrieval-augmented image captioning for world knowledge and specific details.
- [S1]  **daso**: A unified codebase for semi-supervised learning (SSL), related to [C3].
 - Supports training of SSL algorithms, including techniques for class-imbalance.
 - Integrates 6 base SSL algorithms and 4 class-imbalanced methods within a single codebase.

Academic Services

Conference Reviewer

- Conference on Computer Vision and Pattern Recognition (**CVPR**): 2022 – 2025
- European Conference on Computer Vision (**ECCV**): 2022 (**Outstanding**), 2024
- International Conference on Computer Vision (**ICCV**): 2023
- Conference on Neural Information Processing Systems (**NeurIPS**): 2023, 2024 (**Top Reviewer**)
- International Conference on Learning Representations (**ICLR**): 2024, 2025
- Winter Conference on Applications of Computer Vision (**WACV**): 2024, 2025
- International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**): 2024, 2025
- AAAI Conference on Artificial Intelligence (**AAAI**): 2025

Journal Reviewer

- International Journal of Computer Vision (**IJCV**): 2023, 2024

Teaching Experience

Teaching Assistant (TA) at EE, KAIST

- EE735: Computer Vision (Fall, 2020)
- EE898: Advanced Topics in Deep Learning for Robotics and Computer Vision (Spring, 2020)
- EE405: Electronics Design Lab (Fall, 2019)